

Deep Imputation Bi-Stochastic Graph Regularized Matrix Factorization for Clustering Single-Cell RNA-Sequencing Data

Wei Lan^{1b}, Jianwei Chen^{1b}, Mingyang Liu^{1b}, Qingfeng Chen^{1b}, Jin Liu^{1b}, Jianxin Wang^{1b},
and Yi-Ping Phoebe Chen^{1b}

Abstract—By generating massive gene transcriptome data and analyzing transcriptomic variations at the cell level, single-cell RNA-sequencing (scRNA-seq) technology has provided new way to explore cellular heterogeneity and functionality. Clustering scRNA-seq data could discover the hidden diversity and complexity of cell populations, which can aid to the identification of the disease mechanisms and biomarkers. In this paper, a novel method (DSINMF) is presented for clustering single cell RNA sequencing data by using deep matrix factorization. Our proposed method comprises four steps: first, the feature selection is utilized to remove irrelevant features. Then, the dropout imputation is used to handle missing value problem. Further, the dimension reduction is employed to preserve data characteristics and reduce noise effects. Finally, the deep matrix factorization with bi-stochastic graph regularization is used to obtain cluster results from scRNA-seq data. We compare DSINMF with other state-of-the-art algorithms on nine datasets and the results show our method outperforms than other methods. The code can be downloaded from <https://github.com/lanbiolab/DSINMF>.

Index Terms—ScRNA-seq data clustering, deep matrix factorization, bi-stochastic graph regularization.

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Wei Lan is with the School of Computer, Electronic and Information, Guangxi Key Laboratory of Multimedia Communications and Network Technology, Guangxi University, Nanning 530004, China (e-mail: lanwei@gxu.edu.cn).

Jianwei Chen and Mingyang Liu are with the School of Computer, Electronic and Information, Guangxi University, Nanning 530004, China (e-mail: jaychen.oxxx@foxmail.com; hitomil@foxmail.com).

Qingfeng Chen is with the School of Computer, Electronic and Information, State Key Laboratory for Conservation and Utilization of Subtropical Agro-bioresources, Guangxi University, Nanning 530004, China (e-mail: qingfeng@gxu.edu.cn).

Jin Liu and Jianxin Wang are with the Hunan Provincial Key Lab on Bioinformatics, School of Computer Science and Engineering, Central South University, Changsha 410083, China (e-mail: liujin06@mail.csu.edu.cn; jxwang@mail.csu.edu.cn).

Yi-Ping Phoebe Chen is with the Department of Computer Science and Information Technology, La Trobe University, Melbourne, VIC 3086, Australia (e-mail: phoebe.chen@latrobe.edu.au).

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I. INTRODUCTION

SCRNA-SEQ provides a forceful strategy to reveal those unknown heterogeneity and functional diversity of cell populations, which can help to understand the pathogenesis of diseases. Clustering scRNA-seq data is a crucial technique for identifying the dissimilarity of cell types. Identifying cell types from scRNA-seq data is an unsupervised learning task. This task faces some challenges. First, the noise may degrade the performance of many existing methods [1]. Next, the large number of features in scRNA-seq data impair the performance of many similarity-based clustering algorithms. Third, the RNA-seq technology fails to detect low-expression genes, i.e., dropout events [2], [3], [4]. Moreover, the data sparsity also limits the applicability of clustering methods in single-cell data analysis [5], [6].

In recent years, several approaches have been proposed for clustering single cell transcriptome data [7], [8]. These methods can be divided into two categories: methods based on matrix decomposition and methods based on deep learning. The methods based on matrix decomposition methods are utilized to decompose the scRNA-seq data matrix into sub matrix to extract the features. Chen et al. [9] developed a computational method (SNN-Cliq) based on shared nearest neighbor to cluster scRNA-seq data. By combining quasi-cliques, it created an affinity matrix based on nearest neighbor and obtained the cluster result based on affinity matrix. Park et al. [10] proposed an algorithm (MPSSC) to cluster scRNA-seq data by using multiple doubly stochastic affinity matrices to construct a stable similarity matrix. Kiselev et al. [11] developed a method (SC3) for scRNA-seq analysis by applying spectral clustering to the original matrix. In addition, it performed principal component analysis on the resulting submatrices with different dimensions to obtain the clustering results. Wu et al. [12] proposed a nonnegative matrix factorization algorithm (SLNMF) to cluster scRNA-seq data by imposing the structural constraint on structure matrix. Zheng et al. [13] proposed a method (SinNLRR) based on subspace clustering to obtain scRNA-seq data clusters. It measured the similarity between samples by using linear representation. A matrix factorization-based method (DRjCC) for identifying cell types from scRNA-seq data was presented by Wu et al. [14]. These methods are intuitive, easy to understand and implement. However, some approaches fail to extract the features of the data on large-scale scRNA-seq datasets.

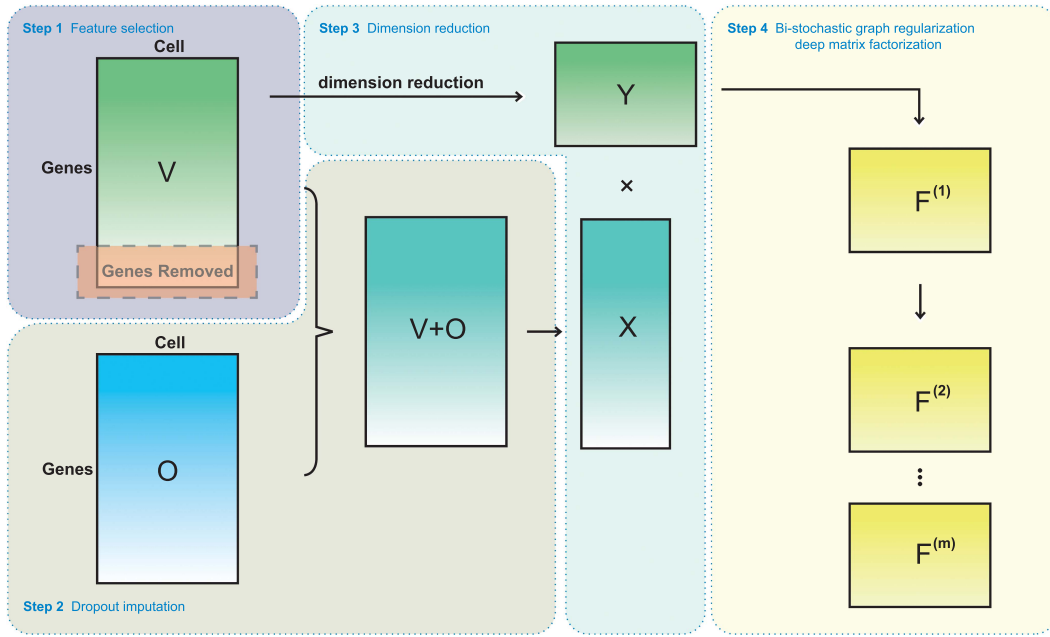


Fig. 1. The flowchart of the DSINMF algorithm. First, a genetic screening method is used to filter out irrelevant genes from the original data matrix V . Second, a matrix factorization technique is applied to impute the missing values in the data matrix and obtain a low-dimensional representation O . Third, a deep matrix factorization model is used to extract the deep features of the data that capture the nonlinear relationships among genes.

The methods based on deep learning can capture hidden features from scRNA-seq datasets and enhance the robustness of the clustering results. Satija et al. [15] introduced a computational method (Seurat) by utilizing the common closest neighbors of network to detect cell types. Wang et al. [16] proposed a clustering approach (SIMLR) based on multiple kernel learning for scRNA-seq analysis. Gedde et al. [17] developed a cluster integration framework based on autoencoder (scCCESS) to integrate different clustering methods for scRNA-seq analysis. Wang et al. [18] proposed a rare cell cluster method (scLDS2) based on deep adversarial learning. Chen et al. [19] proposed a novel and powerful clustering method (scDMFK) that combines the deep autoencoder approach with a statistical model for single-cell transcriptome data. Wang et al. [20] presented an adversarial autoencoder-based algorithm to cluster scRNA-seq data by incorporating adversarial training and semi-supervised method. Wang et al. [21] proposed a deep subspace clustering method (scGDC) to identify cell cluster from scRNA-seq data by learning the topological structure. These methods can handle high-dimensional, large-scale and complex data. But these methods require a large amount of data and computational resources for training. In addition, some approaches ignore the problem of dropout in scRNA-seq, which may result in bias in the classification result.

In this work, we introduce a novel method based on deep matrix factorization (DSINMF) for clustering single-cell RNA-seq data. First, we use the feature selection to remove redundant features from the data, the dropout imputation is used to fill in the missing values. Then, the dimension reduction is used to reduce the noise effects in the data. Finally, the deep matrix factorization with bi-stochastic graph regularization is applied to cluster the scRNA-seq data by extracting the characteristics

from various subspaces. The contributions of this research are as follows:

- To address the challenge of dropout inherent in initial data, matrix imputation is utilized to fill in missing values to address the data sparsity problem.
- Through the utilization of multi-layer matrix factorization structure, our method is able to extract the latent features across different levels.
- We develop a bi-stochastic graph regularization deep matrix factorization method by using the bi-stochastic matrix and the adaptive graph learning to obtain the clustering result from the initial data structure.
- DSINMF updates the partial parameters adaptively, which eliminates the need for manual parameter tuning by the user.

II. METHODS

Fig. 1 illustrates the flowchart of the DSINMF method. It consists of four steps: feature selection, dropout imputation, dimension reduction, and bi-stochastic graph regularization deep matrix factorization.

A. Feature Selection

Clustering scRNA-seq data requires computing the resemblance of cells based on graph regularization constraint. The quality of similarity matrix depends on the reliability of feature information, which affects the clustering performance. The CaFew [22] is utilized to obtain the matrix $H \in \mathbb{R}^{K \times Q}$ which represents the connection strengths between genes and clusters. Here, Q is the number of genes, while K is the number of clusters. We use the coefficient of variation (CV) which is the

ratio of standard deviation to the mean to detect the change of weights. For fitting the relationship between CV and average weight, a linear model is utilized. This linear model is defined as: $\log(CV^2) = a \cdot \log_{10}(\text{mean}(H_p)) + b$. After the fitting step, we can obtain the d_p which is utilized to represent the residuals between the coefficient of variation and the fitted value. Then, we normalized the d_p by using: $Z_p = (d_p - \bar{d}_p)/\sigma_p$, where Z_p represents the relevance of the gene in clustering. The σ_p can reflect the standard deviation of H_p . The value of z_p is inversely correlated with the importance of genes. Therefore, z_p value is utilized as the criterion for feature selection.

B. Dimension Reduction and Dropout Imputation

Given a data matrix, the non-negative matrix factorization (NMF) method aims to find two non-negative matrices that approximate the data matrix. The problem of NMF can be defined as follows:

$$\min_{X,Y} \|V - XY\|^2. \quad (1)$$

The matrix $V \in R^{q \times n}$ represents the gene-by-cell expression data, where the numbers of genes and cells are denoted as q and n , respectively. The Frobenius norm, denoted as $\|\cdot\|$, is used to measure the distance between V and its approximation by two non-negative submatrices $X \in R^{q \times r}$ and $Y \in R^{r \times n}$, where r is the dimension of the subspace.

The raw data has a high dimensionality that renders the spatial distance or correlation measures useless for determining sample groups and measuring cell similarity. Therefore, we use NMF to obtain a low-dimensional representation of the data matrix V . Sparse representation can enhance the readability and accuracy in single-cell transcriptomic data [23]. We use the L1-norm as a regularization term to achieve sparsity, which is defined as

$$\min_{X,Y} \|V - XY\|^2 + \lambda_1 \sum_{i=1}^n \|y_{.i}\|_1. \quad (2)$$

The column vector of Y is denoted by $y_{.i}$, and the L_1 -norm is denoted by $\|\cdot\|_1$. λ_1 is the sparseness penalty parameter. The gene-cell expression matrix has many zero entries due to the dropout phenomenon [24], [25]. To address this issue, we use the matrix $O \in R^{p \times n}$ to account for dropouts, where $O_{ij} = 0$. The matrix $V + O$ represents the original data is the matrix for dropout recovery. It is roughly represented by the product of two matrices X and Y ($V + O \approx XY$). We assume that O is sparse because the matrix has many elements that are equal to zero. As a result, the function is defined as follows:

$$\begin{aligned} \min_{X,Y} & \|V + O - XY\|^2 + \lambda_1 \sum_{i=1}^n \|y_{.i}\|_1 + \lambda_2 \sum_{i=1}^n \|o_{.i}\|_1 \\ \text{s.t. } & O_{ij} \begin{cases} \geq 0, & \text{if } \{i, j\} \in \phi \\ = 0, & \text{if } \{i, j\} \in \phi^* \end{cases}, \\ & 0 \leq i \leq q, 0 \leq j \leq n, Y \geq 0, \\ & \phi = \{(i, j) | V_{ij} = 0, 1 \leq i \leq q, 1 \leq j \leq n\}, \\ & \phi^* = \{(i, j) | V_{ij} > 0, 1 \leq i \leq q, 1 \leq j \leq n\}. \end{aligned} \quad (3)$$

The penalty parameters for sparse control are represented by λ_1 and λ_2 , respectively.

C. Deep Matrix Factorization

Non-negative Matrix Factorization decomposes the matrix V into two smaller matrices X and Y , where X is the feature matrix of data sample and Y is the matrix that uses the coefficients to merge the attributes to reproduce the data. Y also acts as a mapping of the data to a lower dimension. To uncover the latent characteristics of the data, deep matrix factorization is used to extract the deep features [26], [27]. The feature representation matrix C ($C = \prod_{i=1}^m C_i$) contains all the extracted features, and the matrix F is the ratio of the traits when they are combined. It is defined as follows:

$$\begin{aligned} Y & \approx C^{(1)} F^{(1)} \\ F^{(1)} & \approx C^{(2)} F^{(2)} \\ F^{(2)} & \approx C^{(3)} F^{(3)} \\ & \vdots \\ F^{(i)} & \approx C^{(i+1)} F^{(i+1)} \\ \text{s.t. } & C^{(i)} \geq 0, F^{(i)} \geq 0, i = 1, 2, \dots, m, \\ & C^{(i)} \in R^{s_{i-1} \times s_i}, F^{(i)} \in R^{s_i \times n}, k < s_i < r, \\ & 1 < j < m, C^{(1)} \in R^{r \times s_1}, C^{(m)} \in R^{s_{m-1} \times k}, \\ & F^{(1)} \in R^{s_1 \times n}, F^{(m)} \in R^{k \times n}, \end{aligned} \quad (4)$$

where k stands for the quantity of clusters. The entire layer of decomposition is represented by m .

The decomposition of matrix $F^{(i)}$ from the i th layer is used as the input for the next layer. The final matrix $F^{(m)}$ is obtained after the decomposition of m layers. However, NMF cannot preserve the geometric structure of the high-dimensional data that lies on a low-dimensional manifold [28]. Therefore, the projection of the data onto a lower-dimensional space may distort the original structure [26], [29]. To ensure that the similar samples in the original space can remain correlated in the projected space, each layer decomposition should inherit the geometrical arrangement. Therefore, it is defined as

$$\begin{aligned} \min_{C_{i+1}, F_{i+1}} & \|F^{(i)} - C^{(i+1)} F^{(i+1)}\|^2 \\ & + \frac{\beta}{2} \sum_{c,z=1}^n W_{cz} \|F_{.c}^{(i+1)} - F_{.z}^{(i+1)}\|^2 \\ \text{s.t. } & C^{(i+1)} \geq 0, F^{(i+1)} \geq 0, 0 < i < m. \end{aligned} \quad (5)$$

The total number of decomposition layers is denoted by m . The regularization term has a penalty coefficient β . The columns of C and F in the i th row are represented by the vectors $C_{.i}$ and $F_{.i}$, respectively. The matrices $C^{(i)}$ and $F^{(i)}$ are the results of the i th decomposition. The adoption matrix $W \in R^{n \times n}$ is obtained by using the cosine similarity measure, where each element indicates the similarity between two cells. The preservation of the nearby configuration of the geometry can be transformed to a trace optimization problem [30], [31], i.e.,

$$\frac{1}{2} \sum_{c,z=1}^n W_{cz} \|F_{.c} - F_{.z}\|^2 = \text{Tr}(FLF^T)$$

$$s.t. \quad L = D - W, d_i = \sum_{j=1}^n W_{ij}. \quad (6)$$

where the Laplacian matrix of the graph W is denoted by L . The diagonal matrix is denoted by D . Therefore, the (5) can be rewritten as follows:

$$\begin{aligned} \min_{C_{i+1}, F_{i+1}} & \left\| F^{(i)} - C^{(i+1)} F^{(i+1)} \right\|^2 \\ & + \beta \bullet Tr \left(F^{(i+1)} L F^{(i+1)T} \right) \\ s.t. \quad & C^{(i+1)} \geq 0, F^{(i+1)} \geq 0, 0 < i < m. \end{aligned} \quad (7)$$

D. Bi-Stochastic Graph Regularized Matrix Factorization

The similarity matrix cannot be directly used to impose the matrix factorization with the manifold structural constraint. The reason is that the K-NN (K-Nearest Neighbor) based graph may not reflect the initial data structure. Therefore, we employ the bi-stochastic graph regularization [32], [33] to create an adaptive graph Q from the input graph W . The graph Q should have two qualities: first, Q should show the likelihood that any two samples will belong to the same cluster. Therefore, each row of Q should sum up to 1, and each element of Q should be non-negative. Second, Q should be symmetric as it reflects the degree of similarity between each cell pair. Moreover, we add the restriction $diag(Q) = 0$ to ensure that there is no self-connection for any vertex. We solve this following problem to obtain the bi-stochastic matrix:

$$\begin{aligned} \min_Q & \|Q - W\|^2 \\ s.t. \quad & Q \geq 0, Q = Q^T, \sum_{i=1}^n Q_{ij} = 1, diag(Q) = 0. \end{aligned} \quad (8)$$

We combine the (7) and (8) to formulate the following matrix factorization with a bi-stochastic graph constraint:

$$\begin{aligned} \min_{C_{i+1}, F_{i+1}} & \left\| F^{(i)} - C^{(i+1)} F^{(i+1)} \right\|^2 \\ & + \beta \bullet Tr \left(F^{(i+1)} (E - Q) F^{(i+1)T} \right) + \xi \|Q - W\|^2 \\ s.t. \quad & C^{(i+1)} \geq 0, F^{(i+1)} \geq 0, 0 < i < m, Q \geq 0, \\ & Q = Q^T, \sum_{i=1}^n Q_{ij} = 1, diag(Q) = 0. \end{aligned} \quad (9)$$

E. Optimization of DSINMF

There is no convexity in the objective function of this model, so we can use the alternating direction method of multipliers (ADMM) [34] to solve the optimization question.

We split (8) into two smaller problems to solve it

$$\min_P \|Q - W\|^2 \quad s.t. \quad Q = Q^T, \sum_{i=1}^n Q_{ij} = 1, \quad (10)$$

and

$$\min_Q \|Q - W\|^2 \quad s.t. \quad Q \geq 0, diag(Q) = 0. \quad (11)$$

We solve the (10) as follows:

$$Q_1 = G + \frac{n + 1^T G 1}{n^2} \mathbf{1} \mathbf{1}^T - \frac{1}{n} G \mathbf{1} \mathbf{1}^T - \frac{1}{n} \mathbf{1} \mathbf{1}^T G, \quad (12)$$

we define $G = \frac{W + W^T}{2}$ as the symmetric matrix of W , $\mathbf{1}$ as the vector of ones, and n as the dimension of W .

We solve the (11) as follows:

$$Q_2 = \max(W, 0), \quad diag(Q_2) = 0. \quad (13)$$

To obtain Q , we update the variables Q_1 and Q_2 alternately, where Q_1 corresponds to W in (13) and Q_2 corresponds to W in (12).

We define the Lagrange multiplier for the non-negativity constraint on Y as $\Theta \in R^{s_1 \times n}$. The Lagrangian function of the function to optimize for DSINMF is given by

$$\begin{aligned} L(X, Y) = & \|V + O - XY\|^2 \\ & + \lambda_1 \sum_{i=1}^n \|y_{.i}\|_1 + \lambda_2 \sum_{i=1}^n \|o_{.i}\|_1 \\ & + \alpha \bullet Tr \left(F (E - Q) F^T \right) + Tr \left(\Theta Y^T \right). \end{aligned} \quad (14)$$

We compute the partial derivatives of the dependence of the Lagrangian function L on X

$$\frac{\partial L(X, Y)}{\partial X} = -2(V + O - XY) Y^T. \quad (15)$$

By setting $\frac{\partial L}{\partial X} = 0$, we obtain the update rule for X as follows:

$$X \leftarrow X \frac{(V + O) Y^T}{X Y Y^T}. \quad (16)$$

The loss function for the variable Y is given by

$$\begin{aligned} loss(X, Y) = & \|V + O - XY\|^2 + Tr \left(\Theta Y^T \right) \\ & + \lambda_1 \sum_{i=1}^n \|u_{.i}\|_1 \\ s.t. \quad & U - Y = 0. \end{aligned} \quad (17)$$

where the i th column of the matrix V is denoted by $u_{.i}$. ADMM is used to find the optimal solution for (17), which is able to reformulated as follows:

$$\begin{aligned} loss(X, Y; T) = & \|V + O - XY\|^2 + Tr \left(\Theta B^T \right) \\ & + \lambda_1 \sum_{i=1}^n \|u_{.i}\|_1 + \langle T, U - Y \rangle + \sigma \|U - Y\|^2. \end{aligned} \quad (18)$$

where the penalty parameter is denoted by $\sigma > 0$, and the Lagrange multiplier is denoted by $T \in R^{r \times n}$. The slope of the loss function $loss(X, Y; T)$ at a given point with respect to Y is given by

$$\begin{aligned} \frac{\partial loss(X, Y; T)}{\partial Y} = & X^T (V + O) + \sigma E + T \\ & - (X^T X + \sigma E) Y + \Theta. \end{aligned} \quad (19)$$

By applying the condition of Karush-Kuhn-Tucker $\Theta Y^T = 0$, we obtain the update rule for Y by setting $\frac{\partial \text{Loss}}{\partial Y} = 0$ as follows:

$$Y \leftarrow Y \frac{X^T (V + O) + \sigma U + T}{(X^T X + \sigma E) Y}, \quad (20)$$

where E is the matrix with ones on the diagonal and zeros elsewhere. The following formula shows how the variable T is updated:

$$\frac{\partial \text{loss}(X, Y; T)}{\partial T} = U - Y \quad (21)$$

$$T \leftarrow T + \sigma (U - Y). \quad (22)$$

We approximate U by using the (18), and then we obtain

$$U \leftarrow \underset{\sigma}{\text{argmin}} \left\| U - \left(Y - \frac{T}{\sigma} \right) \right\|^2 + \lambda_1 \sum_{i=1}^n \|u_{.i}\|_1. \quad (23)$$

We use the (14) to obtain O , and then we have

$$O \leftarrow \underset{\sigma}{\text{argmin}} \|O - (XY - V)\|^2 + \lambda_2 \sum_{i=1}^n \|o_{.i}\|_1. \quad (24)$$

Eqs. (23) and (24) have the same form as soft threshold problems, our updating scheme of matrix A and matrix B can be expressed as follows:

$$U \leftarrow \text{Soft}_{\frac{\lambda_1}{\sigma}} \left(Y - \frac{T}{\sigma} \right) \quad (25)$$

$$O \leftarrow \text{Soft}_{\lambda_2} (XY - V). \quad (26)$$

where the soft threshold function $\text{Soft}_{\tau}(x)$ is defined as follows:

$$\text{Soft}_{\tau}(x) = \begin{cases} x + \varepsilon, & \text{if } x > \varepsilon \\ x - \varepsilon, & \text{if } x < -\varepsilon \\ 0, & \text{otherwise} \end{cases}. \quad (27)$$

The following equation can be derived for the variable C :

$$\text{loss}(C) = \min_C \|Y - CF\|^2 + \text{Tr}(\Gamma C^T). \quad (28)$$

The function's partial derivatives with respect to C are given by

$$\frac{\partial \text{loss}(C)}{\partial C} = -2(Y - CF)F^T + \Gamma. \quad (29)$$

By applying the condition of Karush-Kuhn-Tucker $\Gamma C^T = 0$, we can obtain the following updating rule for C by setting $\frac{\partial \text{Loss}(C)}{\partial C} = 0$:

$$C \leftarrow C \frac{Y F^T}{C F F^T}. \quad (30)$$

The loss function for the variable F can be derived as follows:

$$\begin{aligned} \text{loss}(F) = \min_C \|Y - CF\|^2 \\ + \beta \bullet \text{Tr}(F(E - Q)F^T) + \text{Tr}(\Theta F^T). \end{aligned} \quad (31)$$

The partial derivatives with respect to F are obtained as follows:

$$\frac{\partial \text{loss}(F)}{\partial F} = -2C^T(Y - CF) + 2\beta F(E - Q) + \Theta. \quad (32)$$

TABLE I
OVERVIEW OF THE DATASETS USED IN EXPERIMENTS

Dataset	Cells	Genes	Types	Species	Ref
Pollen	249	10015	11	Homo sapiens	Pollen et al., 2014
Chu	1018	19097	7	Homo sapiens	Chu et al., 2016
Patel	430	5769	5	Homo sapiens	Patel et al., 2014
Zheng	500	4776	3	Homo sapiens	Zheng et al., 2017
Goolam	124	40315	5	Mus musculus	Goolam et al., 2016
Haber	1522	20108	9	Mus musculus	Haber et al., 2017
Darmanis	420	22085	8	Homo sapiens	Darmanis et al., 2015
Mouse2	1064	14878	13	Mus musculus	Biase et al., 2014
Campbell	20921	33082	20	Mus musculus	Campbell et al., 2017

By applying the KKT condition $\Theta F^T = 0$, we can derive the following updating rule for F by setting $\frac{\partial \text{Loss}(F)}{\partial F} = 0$:

$$F \leftarrow F \frac{C^T Y + \beta F Q}{C^T C F + \beta F}. \quad (33)$$

The following equation shows how we can update (9) with the help of (30) and (33):

$$C^{(i+1)} \leftarrow C^{(i)} \frac{F^{(i)} F^{(i)T}}{C^{(i)} F^{(i)} F^{(i)T}} \quad (34)$$

$$F^{(i+1)} \leftarrow F^{(i)} \frac{C^{(i+1)T} F^{(i)} + \beta F^{(i)} Q}{C^{(i+1)T} C^{(i+1)} F^{(i)} + \beta F^{(i)}}. \quad (35)$$

where the variable F^1 is equal to Y .

F. Datasets and Evaluation Metrics

We collected data from nine different data sources of human and mouse scRNA-seq data for our experiment, comprising five human datasets (Darmanis [35], Pollen [36], Chu [37], Zheng [38], Patel [39]) and four mouse datasets (Goolam [40], Mouse2 [38], Haber [41] and Campbell [42]). We applied log transformation to all original data. Table I summarizes the details of the datasets.

To evaluate the performance of the clustering method, we use three metrics: Adjust Rand Index (ARI) [43], Normalized Mutual Information (NMI) [43] and Adjust Mutual Information (AMI) [44]. These metrics are commonly used to compare the performance between two clustering methods. A higher value of these metrics indicates a better clustering quality.

III. RESULTS

To evaluate the performance of DSINMF, we compare it with nine clustering algorithms including SNN-clique [9], K-means [45], MPSSC [10], NMF [46], pcaReduce [47], Seurat [15], scGMAI [48], scDCC [49] and DRjCC [14].

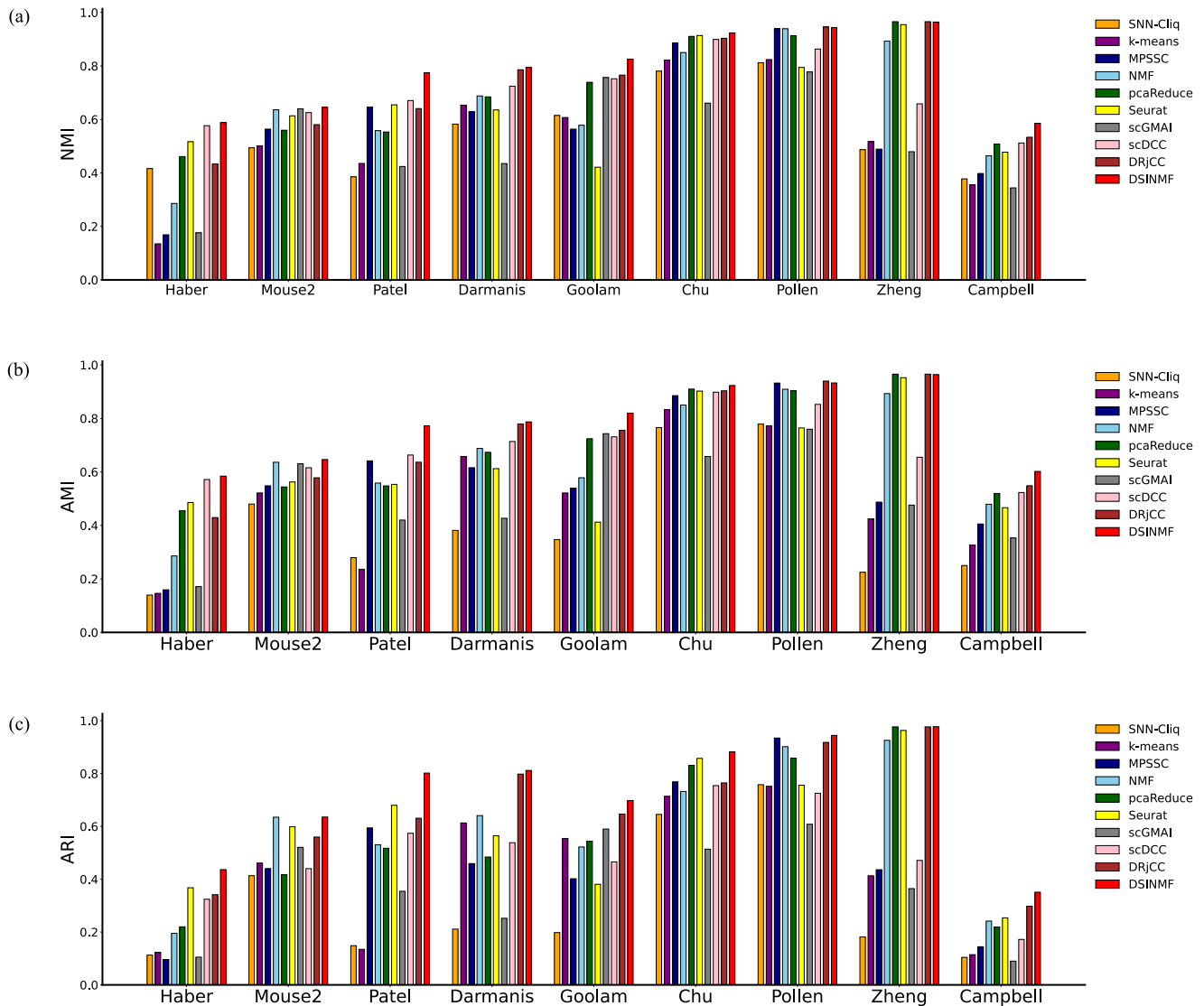


Fig. 2. The performance of different algorithms on nine scRNA-seq datasets in term of three measurements: (A)NMI, (B)AMI and (C)ARI. The numerical results of the experiment are the average of ten operations.

A. The Comparison of Clustering Result

We conduct the experiment on nine scRNA-seq datasets, and the details of each dataset is depicted in Table I. To ensure a fair comparison, we provided the true cluster number to pcaReduce [47], Seurat [15], scGMAI [48], MPSSC [10], scDCC [49] and DRjCC [14], K-means [45], while SNN-Cliq [9] does not allow to set a specific number of clusters. We used the default parameters for all methods. The results of different approaches are summarized in Fig. 2. The DSINMF achieved the best experimental results on seven datasets, and performed comparably with the best method on the other two datasets. As shown in Fig. 2(A), the DSINMF outperformed the other algorithms on seven datasets in terms of NMI. In particular, the DSINMF obtained a 35% improvement (compared to DRjCC) on the Haber dataset. The clustering performance is shown in Fig. 2(B) and (C) in terms of AMI and ARI,

respectively. The DSINMF also achieved great performance on all datasets according to these metrics. In addition, the values of NMI, AMI and ARI of different methods is shown in Table II. In Haber, mouse2, Patel, Darmanis, Goolam, Chu and Campbell, our methods obtained the best performance. In Patel and Goolam, the DSINMF outperformed other methods. The metrics of the second most effective method(scDCC) in Patel are 0.6706(NMI), 0.6631(AMI) and 0.5741(ARI). By using DSINMF, the metrics in Goolam improved to 0.774782(NMI), 0.77208(AMI) and 0.801837(ARI). The metrics of the second most effective method(DRjCC) in Goolam are 0.765769(NMI), 0.7558351(AMI) and 0.646814(ARI). By using DSINMF, the metrics in Goolam improved to 0.82566(NMI), 0.81966(AMI) and 0.69804(ARI).

In Pollen and Zheng, DSINMF rank second in NMI and AMI. The metrics in Pollen using DSINMF are 0.943677(NMI), 0.932666(AMI) and 0.944444(ARI). The metrics in Pollen

TABLE II
CLUSTERING RESULTS ON NINE DATASETS OBTAINED BY 10 METHODS

datasets	Haber			mouse2			Patel			Darmanis			Campbell		
	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI
SNN-Cliq	0.416292	0.139475	0.112757	0.494005	0.479816	0.413452	0.385566	0.279316	0.148399	0.582552	0.381379	0.211146	0.377724	0.249999	0.10432
k-means	0.134983	0.145983	0.123235	0.501406	0.521435	0.461432	0.435541	0.235995	0.134698	0.653546	0.657465	0.613225	0.354748	0.326715	0.114439
MPSSC	0.168774	0.159316	0.096107	0.563853	0.548107	0.440624	0.645612	0.641361	0.594483	0.628973	0.615957	0.459268	0.397959	0.405398	0.144445
NMF	0.285887	0.285887	0.195313	0.636064	0.636064	0.634824	0.558246	0.558246	0.530509	0.687793	0.687793	0.640947	0.464215	0.479033	0.24161
pcaReduce	0.461061	0.455162	0.219821	0.559732	0.543814	0.417723	0.553108	0.547677	0.517297	0.684418	0.673537	0.484155	0.5082	0.519346	0.219121
Seurat	0.516742	0.485634	0.367798	0.613587	0.562458	0.598681	0.654678	0.55343	0.679943	0.635523	0.612340	0.564482	0.477687	0.46622	0.253839
scGMAI	0.176000	0.171043	0.10512	0.639618	0.630609	0.520676	0.424099	0.420006	0.354546	0.434976	0.42686	0.252300	0.344125	0.353202	0.090054
scDCC	0.576600	0.571700	0.324300	0.625600	0.615600	0.440200	0.670600	0.663100	0.574100	0.724500	0.713800	0.538300	0.511921	0.522834	0.172213
DRjCC	0.433721	0.429335	0.341720	0.580570	0.578173	0.559823	0.640629	0.63634	0.630884	0.785715	0.779246	0.797993	0.533499	0.548409	0.298004
DSINMF	0.588212	0.584158	0.436526	0.645297	0.646428	0.635565	0.774782	0.77208	0.801837	0.794362	0.787004	0.811870	0.58563	0.6018	0.35091
datasets	Goolam			Chu			Pollen			Zheng					
	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI			
SNN-Cliq	0.615015	0.347018	0.198109	0.780788	0.765784	0.645550	0.812110	0.77871	0.758082	0.487105	0.224968	0.181605			
k-means	0.607135	0.521435	0.554365	0.822199	0.803243	0.714325	0.823848	0.77243	0.752143	0.516771	0.424335	0.413254			
MPSSC	0.563863	0.539523	0.401974	0.886173	0.885076	0.768858	0.939057	0.932287	0.934521	0.488947	0.487013	0.436111			
NMF	0.578207	0.578207	0.522068	0.850219	0.850219	0.732382	0.939094	0.909095	0.901897	0.892865	0.892865	0.925914			
pcaReduce	0.738952	0.724308	0.544094	0.910638	0.909782	0.831007	0.913178	0.903671	0.858644	0.965638	0.965511	0.976906			
Seurat	0.421359	0.412356	0.380736	0.913728	0.902214	0.857492	0.794566	0.764456	0.756086	0.954367	0.952346	0.963375			
scGMAI	0.756507	0.743002	0.590095	0.661048	0.657660	0.513958	0.777672	0.759686	0.608311	0.47953	0.475914	0.364571			
scDCC	0.752100	0.731200	0.465600	0.899300	0.898100	0.754500	0.863200	0.852600	0.725500	0.658400	0.654900	0.471300			
DRjCC	0.765769	0.7558351	0.646814	0.903605	0.903392	0.764835	0.945661	0.939663	0.917968	0.965674	0.965544	0.976869			
DSINMF	0.825658	0.819661	0.698037	0.923805	0.923211	0.882360	0.943677	0.932666	0.944444	0.964250	0.964119	0.977678			

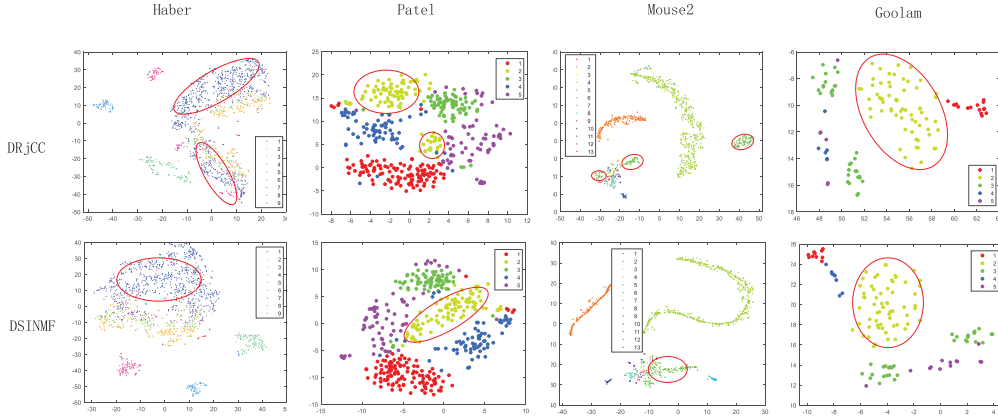


Fig. 3. Visualization of clustering data by DSINMF and DRjCC in Haber, Patel, Mouse2 and Goolam.

using DRjCC are 0.945661(NMI), 0.939663(AMI) and 0.917968(ARI). The metrics in Zheng using DSINMF are 0.964250(NMI), 0.964119(AMI) and 0.977678(ARI). The metrics in Zheng using DRjCC are 0.965674(NMI), 0.965544(AMI) and 0.976869(ARI). The performance of DSINMF and DRjCC in Pollen and Zheng is better than other methods. The possible reason is that the use of dropout imputation in DSINMF may weaken the performance in low complexity data. The Fig. 3 shows the two-dimensional visualization results of DSINMF and DRjCC by using t-SNE on four datasets (Haber, Patel, Mouse2, Goolam). It can be observed that DSINMF can effectively obtain cell cluster with the same category on Haber, Patel and Mouse2 datasets. The cluster distribution obtained by DSINMF is more significant than that obtained by DRjCC.

B. Parameter Analysis

In DSINMF, the parameter λ_1 and λ_2 are the penalty factors of matrices O and B , respectively. In addition, the parameter β is the penalty factors of graph regularization. By adjusting β , we can control the intensity of graph regularization and then affect the relationships between data. Following Shah et al. [46], we update these parameters adaptively during the iteration process.

The update rules are defined as follows:

$$\lambda_1 = \frac{\|V + O\|}{\|Y\|}, \lambda_2 = \frac{\|XY - V\|}{\|O\|} \text{ and } \beta = \frac{\|F_i\|}{\|F_{i+1}\|}. \quad (36)$$

We used the Z_p score to rank the features (genes) according to their importance for distinguishing different cell types. We then selected the top-ranked features based on a percentage threshold. Fig. 4(A) displays the result of this process. It shows that the optimal cluster performance was achieved when selecting features between 50% and 70%.

We also investigated the effect of dimensionality reduction by varying the dimension of the first layer from 80 to 1040. Fig. 4(B) illustrates the result of this experiment. It indicates that the DSINMF obtained the highest NMI score when the dimension was set to 80. Furthermore, we examined the influence of layer number in our method. We set the number of layers from 1 to 4. Table III reports the results of this analysis. It reveals that the best performance was obtained when using four layers matrix factorization.

C. Differentially Expressed Genes Recognition

In scRNA-seq data analysis, differentially expressed genes identification is utilized to identify the differences in gene expression

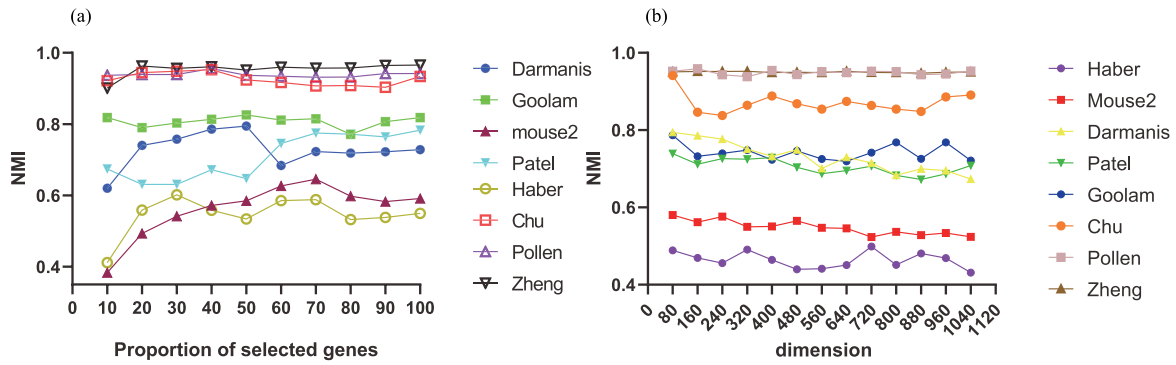


Fig. 4. (A) The impact of feature selection's proportion of selected genes. The horizontal axis shows the percentage of selected genes among the candidate genes. (B) The distinct dimensions' effects on dimension reduction. The horizontal axis shows the target dimension of the dimensionality reduction layer. This figure illustrates how the performance of the eight datasets varies with the target dimension.

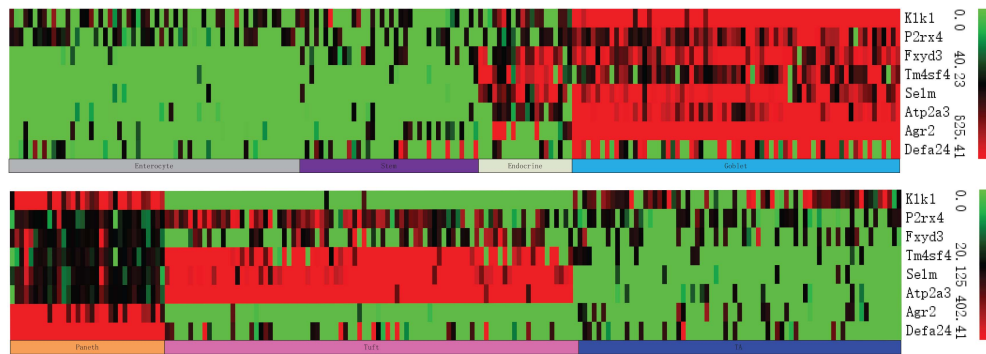


Fig. 5. Heat map of the top eight genes in the Haber dataset, which were selected by our method. The Haber dataset contains the gene expression data of 1,522 cells from mouse epithelial cells, which are classified into seven types. The heat map shows the expression level of each gene in each cell type, with deeper red colors indicating higher expression. The heat map reveals the distinct patterns of gene expression among different cell types, and also demonstrates the validity of our clustering results.

TABLE III
THE PERFORMANCE OF DSINMF IN PATEL WITH DIFFERENT LAYERS

Operating parameters	NMI	AMI	ARI	accuracy
parameter = (80, k)	0.7429348	0.7504162	0.7587987	0.8439535
parameter = (80, 70, k)	0.7633621	0.7703931	0.7799255	0.8613954
parameter = (80,70,60, k)	0.7747828	0.772081333	0.8018376	0.9269303

under different conditions. In this paper, the Kruskal Wallis test [50] is utilized to predict the tags of the gene expression profiles in terms of p-value. We focused on the Haber dataset, which contains 1,522 mouse epithelial cells. We considered a gene as differentially expressed when the p-value is not more than 0.01. It means that its expression in one cluster was significantly higher than in other clusters. Based on this criterion, eight genes with lowest p-value are chosen by DSINMF, and the visualization result is shown in Fig. 5.

It can be found from Fig. 5 that these genes exhibit different expression patterns across different cell types. For example, the expressions of Tm4sf4, Atp2a3, Selm and P2rx4 in Paneth are higher than in Tuft cells, while tAgr2, Defa24 and Klk1 are less expressed in Paneth than in Tuft cells. These contrasting trends suggest that these genes may have different functions in these two cell types. In addition, the sparsity of genes in different

cells is presented in Fig. 6. The sparsity measures how many cells express this gene for given cell type. The figure shows that some genes have high average expression levels but low sparsity, indicating that they are expressed by a few cells with high intensity. Conversely, some genes have low average expression levels but high sparsity, indicating that they are expressed by many cells with low intensity.

We also reviewed some previous studies that reported the biological functions of these genes. According to Gregorieff et al. [51] and Rajapakse et al. [52], Klk1 is a key regulator of kallikrein activity, which affects the differentiation of goblet and Paneth cells. According to Menendez et al. [53], Defa24 encodes an antimicrobial peptide that is produced by mouse Paneth cells. According to Park et al. [54], AGR2 is an important component for synthesis of MUC2. As reported by Vallois et al. [55], Fxyd3 regulates the insulin secretion induced by glucose by affecting the depolarization of the plasma membrane and the influx of Ca^{2+} ions. According to Anderson et al. [56], Tm4sf4 regulates islet progenitor cell differentiation during endocrine development. According to Hadri et al. [57], Atp2a3 produces several alternatively spliced SERCA3 subtypes, which are involved in specific signal transduction pathways in Tuft cells.

TABLE IV
ABLATION EXPERIMENT OF DSINMF WITHOUT DROPOUT IMPUTATION

datasets	Haber			mouse2			Patel			Darmanis			Campbell		
	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI
DSINMF-O	0.505734	0.500006	0.387587	0.593626	0.592926	0.278597	0.729358	0.726106	0.746992	0.461171	0.732674	0.693651	0.58071	0.601398	0.326646
DSINMF	0.588212	0.584158	0.436526	0.645297	0.646428	0.635565	0.774782	0.77208	0.801837	0.794362	0.787004	0.811870	0.58563	0.6018	0.35091
datasets	Goolam			Chu			Pollen			Zheng					
	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI	NMI	AMI	ARI			
DSINMF-O	0.738629	0.728005	0.564456	0.872182	0.871457	0.792998	0.932435	0.92503	0.849348	0.96567	0.96554	0.976869			
DSINMF	0.825658	0.819661	0.698037	0.923805	0.923211	0.882360	0.943677	0.932666	0.944444	0.964250	0.964119	0.977678			

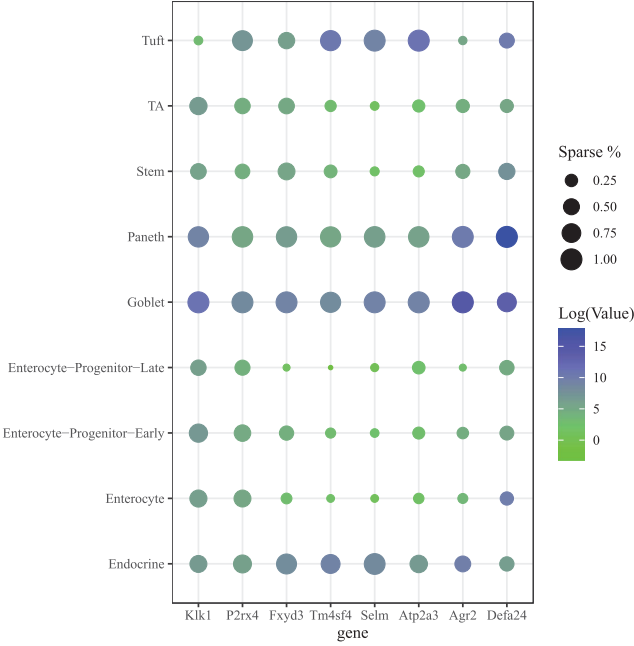


Fig. 6. Top eight gene markers in the Haber dataset, which were identified by our method. The x -axis represents the gene names, and the y -axis represents the cell types. The color intensity indicates the expression level of each gene in each cell type, and the circle size reflects the proportion of cells expressing each gene in each cell type.

D. Ablation Experiment

In order to evaluate the effect of dropout imputation, the ablation experiments are conducted. We remove data imputation matrix to in DSINMF which is represented as DSINMF-O. The result is shown in Table IV. It can be found that the dropout imputation can improve the performance on most datasets. For Zheng dataset, the performance shows a slight decrease in NMI and AMI. The possible reason may be that this dataset have a smaller number of cells and clusters. This suggests that the model can better capture the features of the data and obtain better clustering performance by using dropout imputation.

IV. CONCLUSION

Single-cell RNA-sequencing technology is a new way to explore cellular heterogeneity and functionality. Clustering scRNA-seq can discover the hidden and complexity information. In this paper, we propose deep imputation bi-stochastic graph regularized matrix factorization (DSINMF) for clustering single-cell RNA-sequencing data. In DSINMF, we use four steps for clustering the RNA-seq data: feature selection, dropout

imputation, dimension reduction and deep matrix factorization with bi-stochastic graph regularization. The experimental result show that our method outperforms other state-of-the-art methods. Similar to other methods based on matrix decomposition, our method needs more time cost in dealing with large-scale dataset. As deep learning methods have the ability in feature extraction, we will combine the benefit of matrix factorization and the deep learning to improve scRNA-seq cluster performance in the future [58], [59]. In addition, we also will integrate multi-modal data to enhance robustness [60].

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